Quantitative Trading and Price Impact

End-to-end pipeline with non-parametric impact and multi-day carry backtest

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Modular `src/price_impact/` package | thin orchestration notebook | every backtest run writes to `saved/<name>/`

Pipeline at a glance

Seven assignment sections, eight modules, two enhancements

Modules vs assignment sections

Module	Responsibility	Section
`data`	top-N, σ/ADV	§2.1
`impact_states`	OU $\bar{I}$ (daily + multi)	§2.2/§2.3
`fitting`	OLS + NP + H-grid	§2.2
`alpha`	synthetic α	§2.4
`strategy`	OW/AFS/ext-OW	§2.5
`backtest`	Waelbroeck sim	§2.3
`results`	metrics + TCA + plots	§2.6
`runner`	`BacktestConfig`$\to$disk	§2.6/§2.7

Design principles

- Single-responsibility modules; notebook is glue
- Same simulator runs **any** signed trade path
- Carry mode = one flag; downstream unchanged
- Every run dumps tables + plots to `saved/<name>/`

Two enhancements emphasised: **NP impact** with cross-stock γ shrinkage (§2.2) • **Multi-day carry** backtest engine (§2.3)

Solo submission – all sections by author.

Data preparation

2019 panel, 10-second bins, trailing-20-day daily stats

Panel

- 2019 US equities, 10-s bins (monthly CSVs ∼200 MB each)
- **Top-20 stocks** by full-year total $|q_t|$
- Final panel: ∼11.0M bin rows, 250 trading days

Daily statistics (trailing 20-day rolling)

$$\sigma^2_{i,d} = \overline{\mathrm{Var}_t\, r_{i,d,t}}, \qquad \mathrm{ADV}_{i,d} = \overline{\textstyle\sum_t |q_{i,d,t}|}$$

Used to normalise flow so impact coefficients are comparable across stocks and days.

In-sample / out-of-sample protocol

10 rolling windows:

- train on month m
- tune γ on $m+1$ (NP only)
- validate on $m+2$

for $m = 1, \ldots, 10$.

```
build_panel(data_dir, year,
top_n=20,
   lookback_days=20)
 → PanelData(bins, daily_stats,
stocks)
```

Module `data` = 165 lines, deterministic, no I/O outside the bin CSVs.

Impact-state recursion

OW vs AFS, daily-reset vs multi-day $\bar{I}$

Both models build the OU recursion

$$\bar{I}_{t+1} = (1-\beta)\bar{I}_t + \tilde{q}_t, \qquad \beta = \frac{\ln 2}{H_{\text{bins}}}.$$

They differ only in the kernel applied to raw flow:

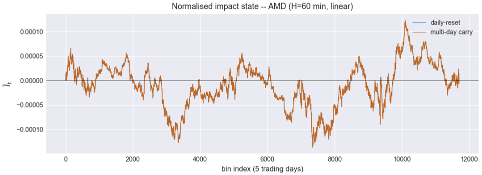
$$\textbf{OW}: \ \tilde{q}_t = \sigma\, q_t / \mathrm{ADV}$$

$$\textbf{AFS}: \ \tilde{q}_t = \sigma\, \mathrm{sgn}(q_t)\sqrt{|q_t|/\mathrm{ADV}}$$

The sqrt form is chosen so the model reproduces the empirical meta-order law $\Delta P/P_0 \propto \sqrt{Q/V}$ (Almgren et al.; Bouchaud): a constant-rate parent of size Q builds equilibrium $\bar{I}^* = f(Q/T)/\beta$, whose shape inherits f. OW would predict linear-in-Q impact, which the data rejects.

Two carry modes produced side-by-side

- `I_bar_daily` – reset at each session
- `I_bar_multi` – carry with overnight decay $(1-\beta)^{16\text{h}\cdot 6}$



*At $H = 60$ min the overnight decay $(1-\beta)^{5760} \approx 10^{-5}$ already wipes $\bar{I}$, so the two curves overlap on screen. The flag still matters for the backtest because **inventory** carries even when $\bar{I}$ does not.*

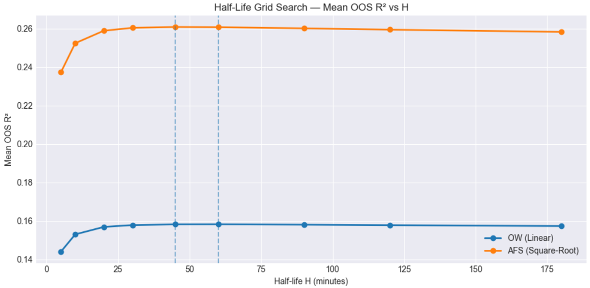
Half-life grid search

Empirical H via rolling-OOS sweep

9-point grid
$H \in \{5, 10, 20, 30, 45, 60, 90, 120, 180\}$ min;
for each (H, model) run rolling OLS and average OOS R^2.

H (min)	OW	AFS
30	0.1579	0.2604
45	0.1582	**0.2609**
60	**0.1583**	0.2608
90	0.1581	0.2602
120	0.1578	0.2595

$H^* = 52.5$ **min** (midpoint).
Cost of universal H^* vs per-model optimum: $< 10^{-4}$.
AFS dominates OW by $\sim$10 **pp** in OOS R^2.



Non-parametric extension + γ shrinkage

ENHANCEMENT #1

Three-sample protocol; method recap

Replace $y = \alpha + \lambda x$ with a binned $g(x)$:

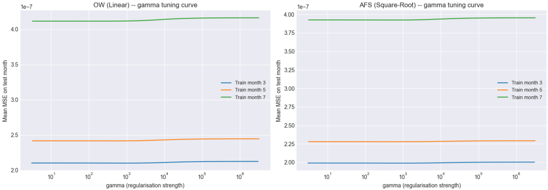
$$g_{i,b}^{\mathrm{reg}}(\gamma) = \frac{S_{y,i,b} + \gamma\,\bar{g}_b}{n_{i,b} + \gamma}$$

- $B = 15$ quantile bins on train month m (pooled across stocks)
- $\bar{g}_b$ = pooled universal curve
- γ **tuned on test month** $m+1$ (MSE), validated on $m+2$

Why three samples?

Tuning γ on the validation month leaks. We need *train, tune, validate* – distinct months, in order.

```
rolling_nonparametric(features,
  n_bins=15, n_windows=10)
 → (summary DF, fits dict)
```



Tuning curves flat across 6 orders of magnitude $\Rightarrow$ in the headline configuration (median $n \approx 3000$ per (stock, bin)) the per-stock estimator is already precise; shrinkage has nothing to fix. We'll see in slide 12 that the estimator behaves correctly when it does need to fix something.

Headline OOS R^2 §2.2

Apples-to-apples comparison + estimated impact curves

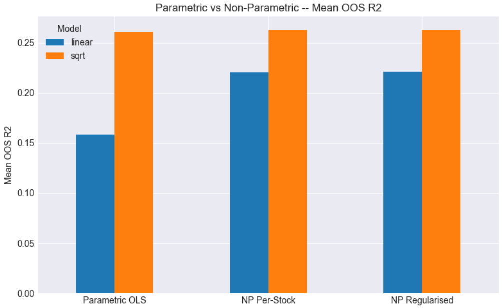
Pooled OOS R^2 (panel-wide null, same merged subset)

Model	Param.	NP Reg.	Δ
OW	0.1612	**0.2206**	$+0.059$
AFS	0.2689	0.2629	-0.006

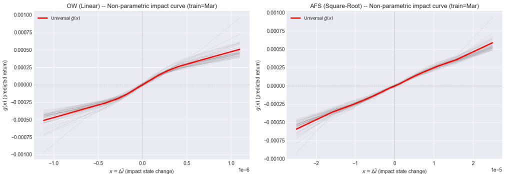
Two findings

- **NP lifts OW by $+5.9$ pp** – linearity is a binding constraint
- **NP ties AFS** (-0.6 pp): the 1-parameter sqrt form sits on the efficient frontier; 15 parameters buy nothing

Aggregation matters: under per-stock R^2 averaging, NP appears to beat AFS by $+0.2$ pp – an artefact that disappears under pooled R^2.



OOS R^2: parametric OLS vs NP estimators



Waelbroeck + carry modes + generic-trade interface

$$P_{\text{sim}}(t) = P_0\big(1 + \mathrm{cumret}(t) + g_{\text{sim}}(t) - g_{\text{ref}}(t)\big), \quad g(t) = \lambda \bar{I}(t) \text{ or } \lambda(t)\bar{I}(t)$$

$\bar{I}^{\text{ref}}$ uses public-tape `orderFlow`; $\bar{I}^{\text{sim}}$ uses the simulator's input trade path.

Generic trades. Engine accepts any signed schedule via a `trade_provider` callable:

- `make_optimal_provider(strategy_fn)` – OW/AFS
- `make_fixed_provider(trades)` – TWAP/VWAP/manual

Carry modes.

- `daily` – reset $\bar{I}$ + inventory each session (baseline)
- `multi` – persist across sessions, overnight decay $(1-\beta)^{16\text{h}\cdot 6}$

Stock splits. Heuristic detector flags $|\Delta p|$, $|\Delta v|$ large + opposite-signed with stable notional. On 2019 top-20: **no splits detected**; engine accepts `exclude_stocks` for hard-coded exclusion.

```
run_backtest(merged, model, trade_provider,
    daily_stats, carry='daily'|'multi',
    overnight_minutes=16*60)
  → BacktestResult
```

Synthetic alpha (§2.4)

$$\alpha_t = r_t^h + y\,\frac{dW_t}{P_t}, \quad y = \sqrt{(\rho^{-2}-1)\,\frac{\mathrm{Var}(r^h)}{E[P^{-2}]h}}$$

Choosing $x=1$ in $\alpha = x\,r^h + y\,dW/P$ gives:

- **Unbiased**: $E[r^h \mid \alpha_t] = \alpha_t$
- **Target correlation**: $\mathrm{Corr}(\alpha, r^h) = \rho$

We use $\rho = 0.05$, $h = 1$ bin (10 s); empirical ρ within $\pm 1\%$ of target.

OW closed form (§2.5)

$$X_t^* = \frac{\alpha_t\,\mathrm{ADV}}{2\lambda\sigma}, \quad \kappa = \beta = \frac{\ln 2}{H_{\text{bins}}}$$

Position adjusts to X^* at speed κ; ramped to zero over the final 30 min.

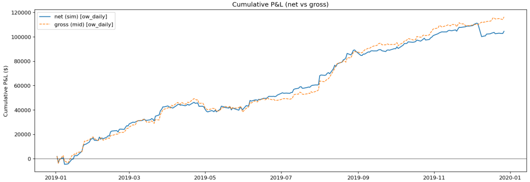
AFS variant. Same target rule per `project.ipynb`; sqrt nonlinearity enters via the simulator's $\tilde{q}$.

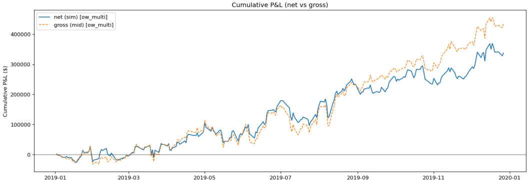
Ext-OW with $\lambda(t)$. Placeholder. Simulator already accepts a per-bin λ_t array; HJB-optimal trade rule is not κ-constant when λ varies – closed form pending.

Position trajectory sketch: X_t rises toward X_t^* at exponential speed κ, clips at $\pm 0.5\%\,\mathrm{ADV}$, ramps linearly to zero in the last 30 min. OW and AFS positions coincide; the cost accounting differs.

Performance – cumulative P&L

Daily-reset vs multi-day, OW model





`ow_daily`: **Sharpe +4.30**, \$105k net, \$12k impact

`ow_multi`: Sharpe +1.61, **\$338k net**, \$98k impact

Net P&L (blue solid) marks to the Waelbroeck-distorted P_{sim}; gross P&L (orange dashed) marks to the undistorted mid.

- **Daily-reset wins on Sharpe** – consistent recycling, tight drawdowns ($-$\$11k vs $-$\$83k)
- **Multi-day wins on dollars** – $\sim 3\times$ net P&L from accumulated inventory, but $\sim 8\times$ impact cost
- Turnover essentially identical between carry modes (\$84M) – only the *cost attribution* changes

TCA decomposition (totals)

Net = Gross − Impact, with α rewards separated

Config	Gross P&L	Impact	Net P&L	Pred. α	Real. α	Turnover
`ow_daily`	116,311	11,787	104,524	44.6 M	116,311	84.0 M
`ow_multi`	435,399	97,813	337,586	44.6 M	435,399	84.0 M
`afs_daily`	119,205	56,417	62,787	44.6 M	119,205	86.6 M
`afs_multi`	482,557	129,999	352,558	44.6 M	482,557	86.6 M

- **Impact cost = Gross − Net** – well-defined under Waelbroeck construction
- **Realised α reward** $\sum_t \mathrm{fwd\_ret}_t \cdot X_t \cdot P_t \equiv$ Gross P&L by construction
- **Predicted α reward** $\sum_t \alpha_t \cdot q_t \cdot P_t$ – dominated by α's noise component at $\rho = 0.05$ (only $\rho^2 = 0.25\%$ of $\mathrm{Var}(\alpha)$ is signal); informative *across* configs, not for level
- Slippage components needing fill-price detail (arrival vs VWAP) intentionally omitted

Live sensitivity sweeps (OW, daily-reset)

Scenario	Sharpe	Net P&L
Baseline	**+4.30**	\$105k
Wrong model	**−3.48**	**−\$863k**
Signal delayed 1 min	**−0.32**	−\$8k
$\rho = 0.01$	+0.45	\$11k
$\rho = 0.10$	**+9.22**	\$251k
$H = 30$ min	+5.91	\$205k
$H = 120$ min	+3.26	\$53k

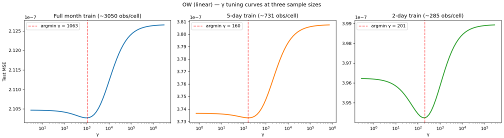
Wrong model is catastrophic (negative Sharpe); **1-min delay flips the sign**; ρ scales linearly; H is forgiving.

Fitting-side stress – γ regime-correctness

Shrink train window 21d → 5d → 2d, per-cell n drops 10×.

γ^*/n: 0.16 → 0.56 → **0.93**.

OW gain (reg−raw): 0.0005 → 0.0070 (∼14×).



Headline flatness is data-rich behaviour, not a defect: the estimator correctly shrinks toward universal when starved.

What we learned

Re-stating the two enhancements with one bullet each

Enhancement 1 – Non-parametric impact with γ regularisation (§2.2)

Apples-to-apples pooled OOS R^2: **OW** $0.161 \to 0.221$ (+5.9 pp, linearity binding); **AFS unchanged at** 0.263 (square-root on the efficient frontier; NP buys nothing). γ shrinkage immaterial in the data-rich headline regime; γ^*/n **rises** $6\times$ **and OOS gain** $14\times$ as the training window shrinks $10\times$ – the estimator behaves correctly under stress.

Enhancement 2 – Multi-day carry backtest (§2.3)

Carrying $\bar{I}$ and inventory across sessions makes the cost of impact *compound*: in the OW config, **impact cost** $\$12$**k** $\to$ $\$98$**k** ($\sim 8\times$), gross P&L $3\times$ **larger**. Daily-reset wins on Sharpe (+4.30 vs +1.61); multi-day wins on absolute P&L. Engine handles both via a single flag; downstream code, TCA columns, and plotting are unchanged.

Honest limitations*: overnight α (§2.4 to-expand) and extended-OW $\lambda(t)$ closed form (§2.5 to-expand) are flagged but not implemented; the simulator side of $\lambda(t)$ is already plumbed via `ImpactModel.lam_t_lookup`.*

Thank you

Questions?

Code: `src/price_impact/` | Notebook: `final_pipeline.ipynb` | Runs: `saved/<name>/`